Software Project Management

Activity Duration (Time) Estimating

Introduction

- Controlling factors in duration estimating
  - ***Historical data***. If available, enterprise- or industry-specific historical data can be used as starting point for estimates
  - ***Activity resource requirements***. The type and number of resources applied to an activity must account for
    - » Resource qualifications
    - » Quantity-quality productivity trade-offs: putting more people on a task may reduce productivity
  - ***Resource calendars***. Must account for full/part time commitments

Sidebar: A useful rule of thumb

- Keep all times in the same units
- Don't mix increments
  - ***Example:*** Schedule everything in hours, then convert to days and fractions as a last step
  - ***Note:*** Most PM software tools enforce this rule by using a uniform time unit and increment

Estimating tools

- ***Expert judgment***. As in resource estimating, one of the most effective tools for estimating. May be used in conjunction with other tools
- ***Analogous estimating*.** Estimation based on similar activities performed in a previous project
  - Most useful if both activity and project are more similar than different
  - Document similarities *and* differences
  - *Example:* Developing the database schema for a project in the same domain and of similar scope

Estimating tools

- ***Reserve analysis*.** Contingency or time reserves can be added to schedule to allow for schedule risks
  - *Contingency reserves* may be used or adjusted as more precise project information becomes available
  - Aka *buffer time*. Do not confuse with slack time.
- ***Three-point estimating techniques*** use a mathematical combination of three different estimates: *Most likely, Optimistic, and Pessimistic*
  - There are variations in defining each of these estimates among different techniques

Three Point Estimating

- Used whenever there is a potential variance in an estimate. Assumes a Gaussian distribution of estimates.
  - Duration
  - Effort
  - Size
- ***Three-point estimating techniques*** use a mathematical combination of three different estimates:
  - Get estimates from experts: Most likely, Optimistic, and Pessimistic
  - *Compute an average:*
    (Optimistic + 4(Likely)+ Pessimistic)/6

Example: Estimation of LOC

CAD program to represent mechanical parts

- Estimated LOC = (Optimistic + 4(Likely)+ Pessimistic)/6

Major Software Functions	ID	Optimistic	Most Likely	Pessimistic	Estimated LOC
User interface and control facilities	UICF	1,500	2,300	3,100	2,300
Two-dimensional geometric analysis	2DGA	3,800	5,200	7,200	5,300
Three-dimensional geometric analysis	3DGA	4,600	6,900	8,600	6,800
Database management	DBM	1,600	3,500	4,500	3,350
Computer graphics display features	CGDF	3,700	5,000	6,000	4,950
Peripheral control	PC	1,400	2,200	2,400	2,100
Design analysis modules	DAM	7,200	8,300	10,000	8,400
Estimated lines of code		23,800	33,400	41,800	33,200

Network Analysis
The First Steps

Network Analysis

- *Network analysis* is the technique that generates the project schedule
- Network analysis may use several different analysis methods to calculate the early and late start dates for project activities. These methods may be combined and include:
  - Gantt Charts
  - Critical Path Method (CPM)
  - Critical Chain Method (CCM)
  - What-if analysis
  - Resource leveling

Network Analysis

- There are a number of network analysis techniques available:
- the ***Critical Path Method*** **(CPM)** to identify the **critical Path**
- Another network analysis technique, the **PERT estimates** for activity duration
- The **Precedence Diagram Method (PDM)** is a graphical network technique that establishes **activity sequencing**
- Network analysis may also make use of mandatory or discretionary parallelism in project activities to allow schedule compression

Network Analysis Terminology

- **Activity.** An activity always consumes time and may also consume resources. I use task and activity equivalently
- **Critical.** A critical activity or event is one that must be achieved by a certain time, having no latitude (slack or float)
- **Critical path.** The critical path is the longest path through a project network. Because it has no slack, all activities on the critical path must be completed as scheduled, or the end date will slip
- **Events.** Beginning and ending points of activities are known as events. An event is a specific point in time
- **Milestone.** An event representing a point in a project of special significance. Usually the completion of a major phase of the work. Project reviews are often conducted at milestones

Creating a precedence table

- A **Precedence table** documents task durations and interdependencies
- Estimating duration of each task
  - Duration estimated from historical information:
    - » Ideally, based on organization's historical experience, if available
    - » More likely: ‘expert’ knowledge
  - Estimation done by task leaders or functional managers
  - Any potentially risky task (technology, skills, dependencies) should include contingency factor
  - Include contingencies to mitigate potential resource shortages

Creating a precedence table

- Determining task interdependencies
  - Goal is to determine predecessor/successor relations
  - Understanding interdependencies allows proper ordering in scheduling tasks
  - Understanding interdependencies also helps in finding possible parallel tasks, which can shorten schedule
  - Parallel tasks should be *truly* independent to minimize risk of backtracking
  - Determining task interdependencies must be a team effort to avoid unpleasant surprises

Sample evolutionary precedence table

	WBS	Task Name	Duration	Predecessors
1	**1**	**- Automatic Toll System**	**65 days?**	
2	**1.1**	**+ Management**	**1 day?**	
7	**1.2**	**- Environment**	**10 days?**	
8	1.2.1	Inception phase environment	10 days?	
9	1.2.2	Elaboration phase environment	10 days?	
10	1.2.3	Construction phase environment	5 days?	
11	1.2.4	Transition phase environment	1 day?	
12	**1.3**	**+ Requirements**	**1 day?**	
17	**1.4**	**- Design**	**40 days?**	
18	1.4.1	Inception phase architecture prototyping	10 days?	8
19	**1.4.2**	**- Elaboration phase design**	**16.67 days?**	
20	1.4.2.1	Architecture design	10 days?	9
21	1.4.2.2	Preliminary component design	6.67 days?	20
22	**1.4.3**	**- Construction phase design modeling**	**20 days?**	
23	1.4.3.1	Architecture design maintenance	20 days?	20,10
24	1.4.3.2	Component design	6.67 days?	21
25	1.4.4	Transition phase architecture design maintenance	1 day?	
26	**1.5**	**- Implementation**	**65 days?**	
27	1.5.1	Inception phase implementation	15 days?	18
28	1.5.2	Elaboration phase implementation	15 days?	21,27
29	1.5.3	Construction phase implementation	15 days?	24,28
30	1.5.4	Transition phase implementation	1 day?	
31	**1.6**	**+ Assessment**	**1 day?**	
36	**1.7**	**+ Deployment**	**1 day?**	

Applying leads and lags

- *Applying leads and lags* allows refinement of a schedule once the major schedule network analysis effort has been completed
- Use of leads and lags may be used to meet imposed constraints, help in resource leveling, or incorporate contingency reserves into a schedule

Calendars

- Identify days and dates when work can be performed
- Affect all project-related activities
- ***General project calendars*** govern overall limitations on when project work may be performed
  - *Example:* Work is performed at a client site, and the **client shuts down for three weeks** during the summer
- ***Resource calendars*** govern limitations on when particular resources (or resource groups) may perform project work
  - *Example:* Individual project team member **vacation schedules**
  - *Example:* Development team **training schedules**

Schedule development output

- **Project schedule network diagrams**
  - Show both project network logic (sequencing) as well as critical path schedule activities
  - Usually displayed as an activity-on-node diagram
- **Gantt charts**
  - Specialized bar charts format to show activity start and end dates, along with durations
  - Easy to read but limited by low information density
- **Milestone charts**
  - ‘Stripped-down’ version of Gantt chart, showing only milestones

Reducing Project Duration

- How can you shorten the schedule?
- Via
  - Reducing scope (or quality)
  - Adding resources
  - Concurrency (perform tasks in parallel)
  - Substitution of activities

Schedule compression

- Shortens the project schedule without changing the project scope, to meet schedule constraints, imposed dates, or other schedule objectives
- There are two types of schedule compression, ***crashing*** and ***fast-tracking***
- ***Crashing*****.** Analyzes cost and schedule trade-offs to get the greatest amount of compression with the least cost
  - *Examples:* Use of **additional resources**, being more efficient, changing approach used to perform work, work overtime
  - Schedule crashing only works for activities where additional resources may shorten the duration of the activity

Schedule compression

- ***Fast-tracking***. Activities that would normally be done **sequentially are done in parallel**.
  - Fast-tracking can lead to rework and increased risks due to unforeseen dependencies
  - Fast-tracking only works when activities can be overlapped to shorten the total duration

Compression Techniques

Shorten the overall duration of the project

- **Crashing**
  - » Looks at cost and schedule tradeoffs
  - » Gain greatest compression with least cost
  - » Add resources to critical path tasks
  - » Limit or reduce requirements (scope)
  - » Changing the sequence of tasks
- **Fast Tracking**
  - » Overlapping of phases, activities or tasks that would otherwise be sequential
  - » Involves some risk
  - » May cause rework
- Barry Boehm says you cannot compress more than 25%

Task

- Name
- ID
- Description of work
- Duration (days)
  - Start Date (Earliest, Latest)
  - Finish Date (Earliest, Latest)
- Resources (People and equipment)
  - Effort (In staff-days)
- Predecessors (other tasks)
- Inputs (documents, decisions, information)
- Successors (other tasks)
- Outputs (documents, decisions, information)

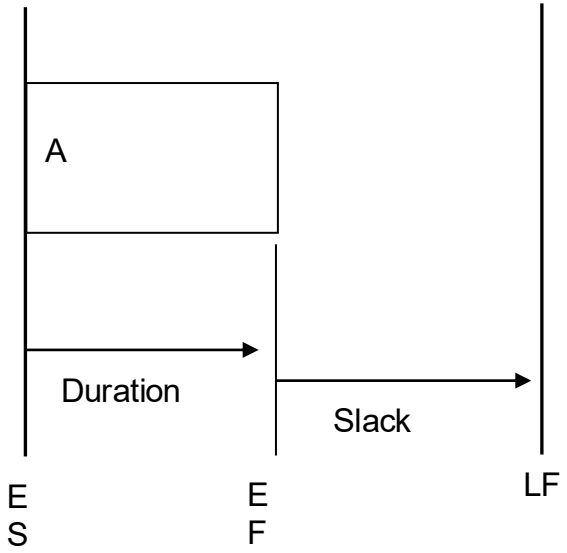
Planning
Project Time Management

CPM: Critical Path Method

PERT: Program Evaluation & Review Technique

Slack Time

- Slack time, also known as float, is the amount of delay expressed in units of time that could be tolerated in the starting time or completion time of an activity without causing a delay in the completion of the project.
- Slack time is the difference between the **late finish** and the **early finish** (LF-EF). If the result is greater than zero, then the activity has a range of time in which it can start and finish without delaying the project completion date, as shown in the figure below:



Slack Time

- If an activity has **zero slack**, it determines the project completion date. In other words, all the activities on the critical path must be done on their earliest schedule or the project completion date will suffer.
- If an activity with total slack **greater than zero** were to be delayed beyond its late finish date, it would become a critical path activity and cause the completion date to be delayed.
- The sequence of activities that has zero slack is defined as the critical path
- In general, the critical path is the path that has minimum slack.

Critical path method

- The *critical path method* (CPM) focuses on calculating theoretical start and finish dates for every activity in the project. In the context of the CPM, the following definitions are essential:
  - *Critical*. A critical activity or event is one that must be achieved by a certain time; a critical activity has no latitude (*slack* or *float*)
  - *Critical path*. The critical path is the longest path through a project network. Because it has no slack, all activities on the critical path must be completed as scheduled, or the end date will slip
- CPM performs these calculation without regard for resource limitations, which can lead to resource over-allocation or inefficient multitasking

Critical path method

- A *forward pass* analysis performs schedule calculations that identify the early start and finish dates of activities and the project
- A *backward pass* analysis performs schedule calculations that identify the late start and finish dates of activities and the project, as well as total and free float
- *Total float* (TF) is the amount of time an activity can be delayed without delaying the project as a whole
- *Free float* (FF) is the amount of time an activity can be delayed without delaying its successor (dependent) activities
- Note that total float is global to the project, while free float is local to the neighborhood of the activity

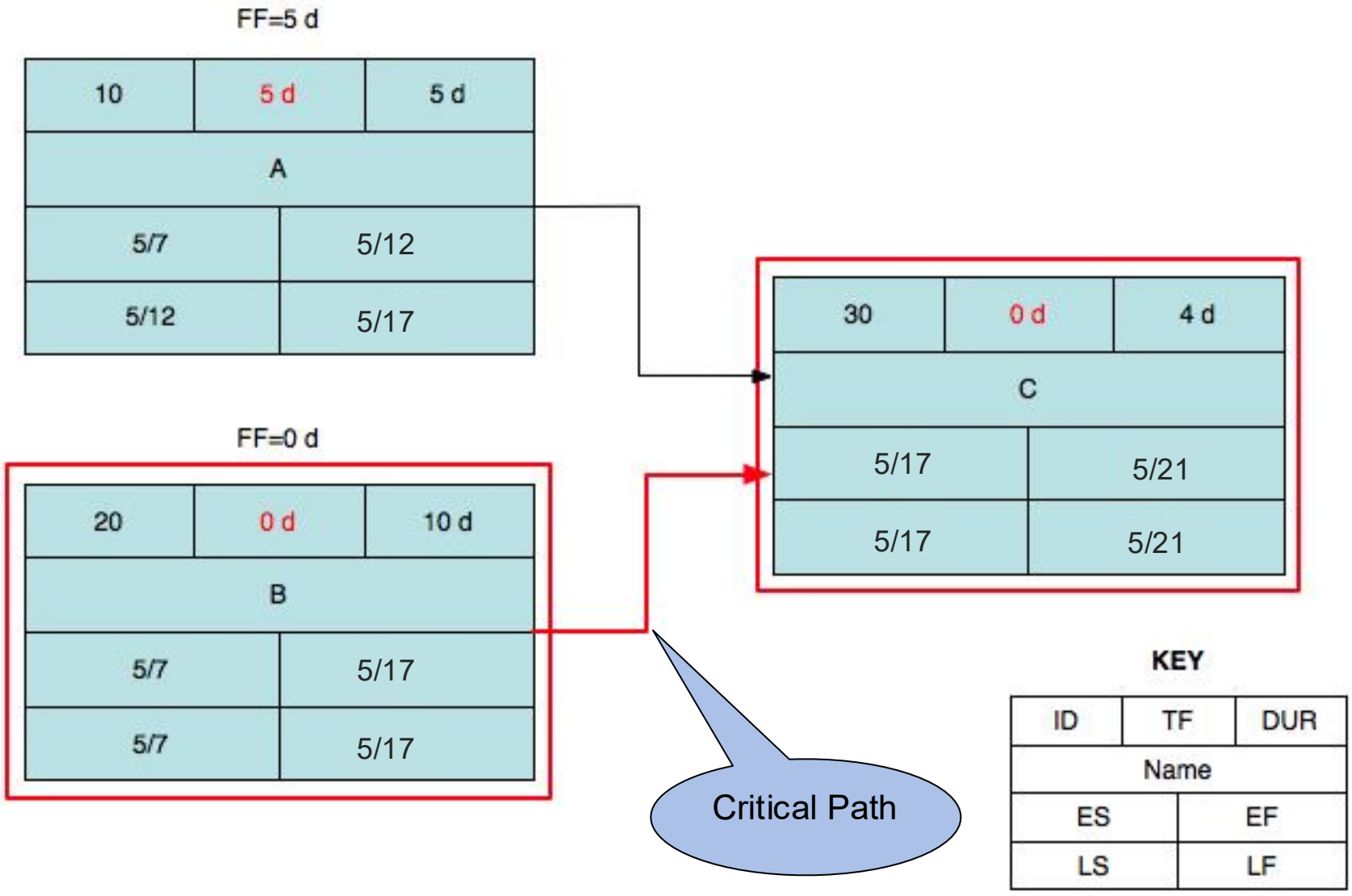
What-if analysis

- Scenario-based analysis of schedule to determine effects of various scenarios on different aspects of the project
  - *Example:* Delay delivery of a critical component by various amounts to determine effect on schedule
  - *Example:* COTS (Commercial Off-The-Shelf) supplier is unable to provide a critical component at all
- What-if scenario analysis effectively tests the robustness of the project schedule in response to adverse circumstances

Resource leveling

- Uses a number of different approaches, including:
  - Assign under-allocated resources to multiple tasks to keep them busy
  - Move key resources off of non-critical tasks
  - Delay start of task until required resources are available
  - Split tasks into two or more subtasks so the subtasks can be assigned to different resources

Simple critical path example



Critical chain method

- The *critical chain method* (CCM) focuses on the resources required for project activities, attempting to keep them leveled throughout the project
- CPM scheduling is rigid and brittle: most activities have little or no float, while the critical path, by definition, has no float whatsoever— any delay in a critical path activity leads to a project delay

Critical path method

- A *forward pass* performs schedule calculations that identify the early start and finish dates of tasks and the project
- A *backward pass* performs schedule calculations that identify the late start and finish dates of tasks and the project, as well as total and free float

Critical path method

- A *forward pass analysis* performs schedule calculations that identify the early start and finish dates of activities and the project
  - *Early Start Date* (ES). ES represents the theoretically earliest date a activity can start
    - ES = Maximum EF of predecessor activity(-ies)
  - *Early Finish Date (EF).* EF represents the theoretically earliest date a activity can finish
    - EF = ES + duration of activity

Critical path method

- A *backward pass analysis* performs schedule calculations that identify the late start and finish dates of activities and the project, as well as total and free float
  - *Late Start Date (LS).* LS represents the theoretically latest date a activity can start without delaying the project
    - LS = LF – duration of activity
  - *Late Finish Date (LF).* LF represents the theoretically latest date a activity can finish without delaying the project
    - LF = Minimum LS of successor activity(-ies)

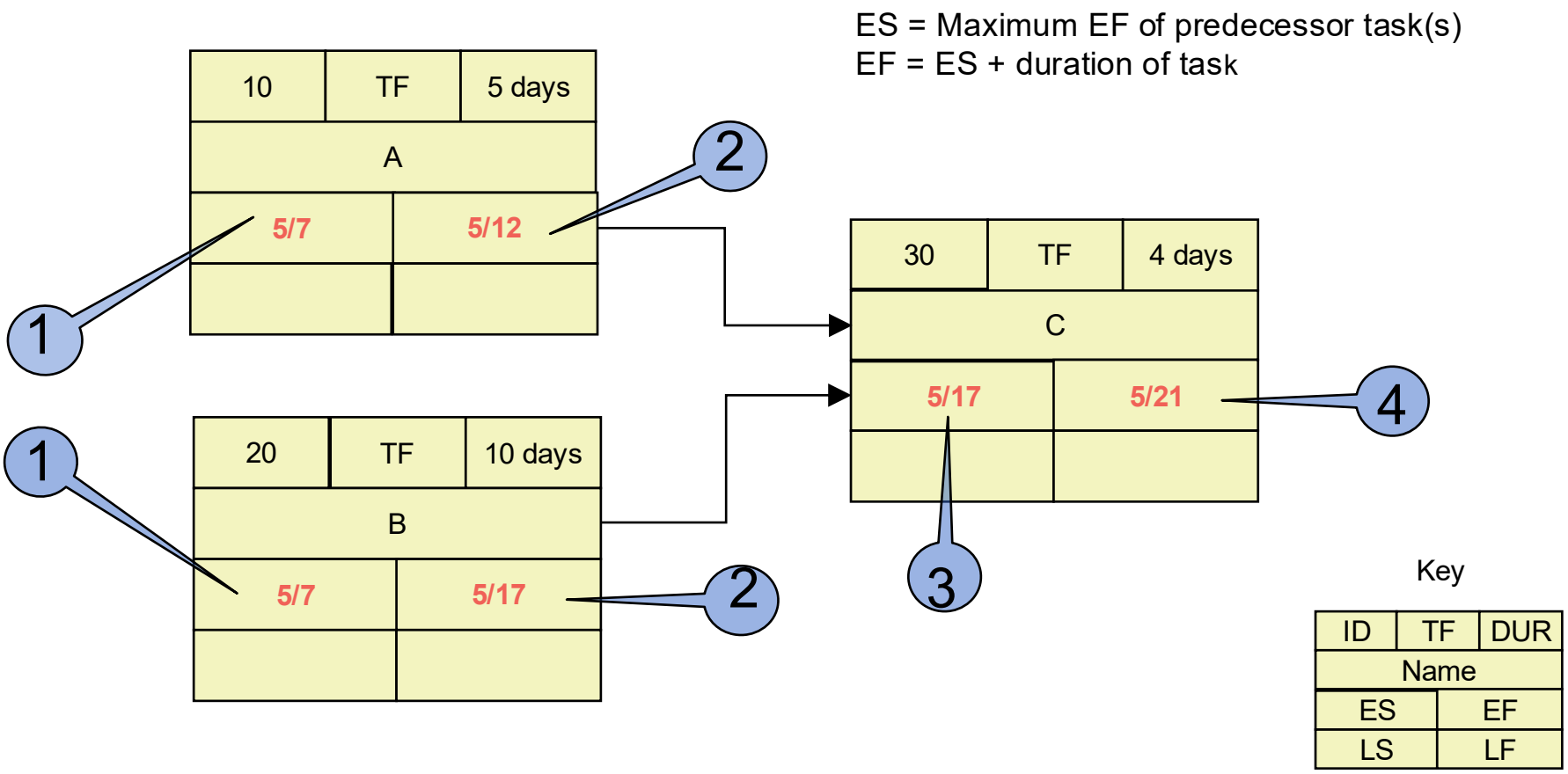
Critical path method

- *Total float (TF)* is the amount of time a activity can be delayed without delaying the project *as a whole*
  - TF = LF - EF
  - If LF < EF then TF < 0
  - If a project has a fixed finish date constraint then TF might be less than zero, meaning it must complete *before* LF to satisfy the finish date constraint
- *Free float (FF)* is the amount of time a activity can be delayed without delaying its successor (dependent) activities
  - FF = Minimum ES (successor activities) – EF
- Note that total float is *global* to the project, while free float is *local* to the neighborhood of the activity

Critical Path Method

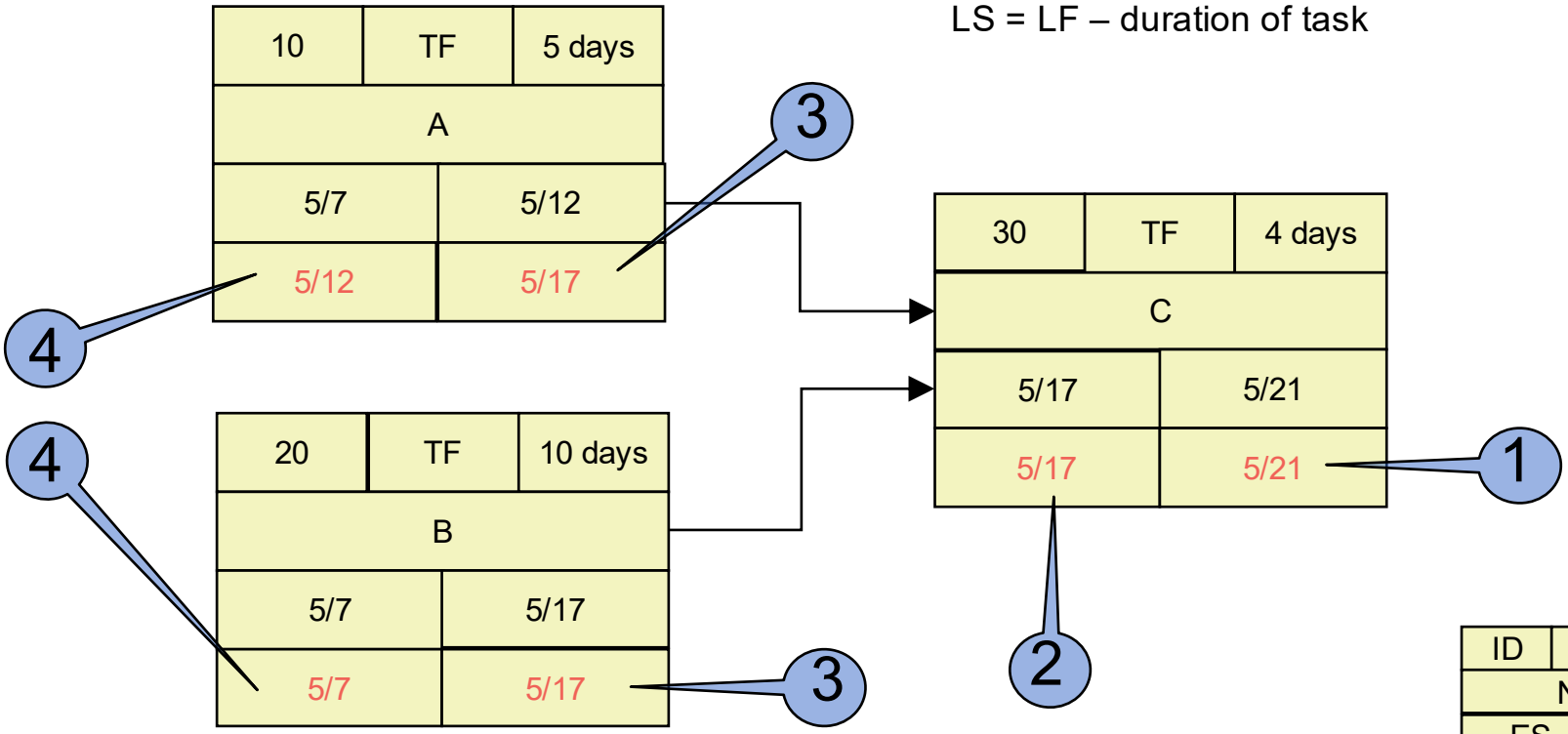
- List all activities in plan.
- Plot tasks onto chart. (Tasks = arrows. End Tasks = dots)
- Show dependencies.
- Schedule activities
  - Sequential activities on critical path.
  - Parallel activities.
  - Slack time for hold-ups.
- Find longest path through chart.
  - This is the critical path
- What is the difference between critical path and critical chain?
  - Critical chain also manages buffer activity durations and resources

Performing forward pass CP analysis

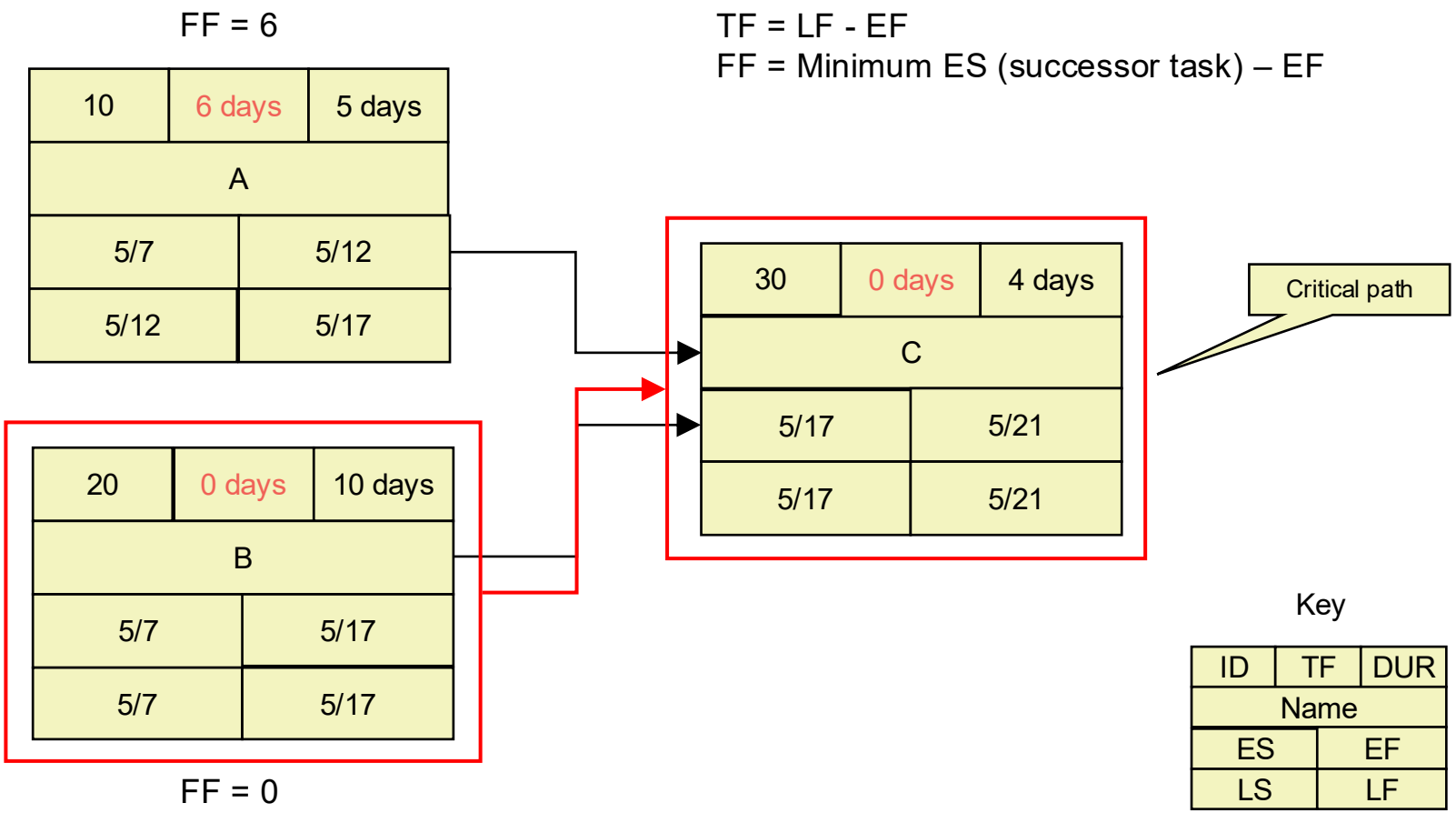


Performing backward pass CP analysis

LF = Minimum LS of successor task(s)
LS = LF – duration of task



Calculating float



Critical Path Method

ACTIVITY	IPA *	DURATION **
A	-	5
B	A	8
C	A	6
D	B	9
E	B, C	6
F	C	3
G	D, E, F	1

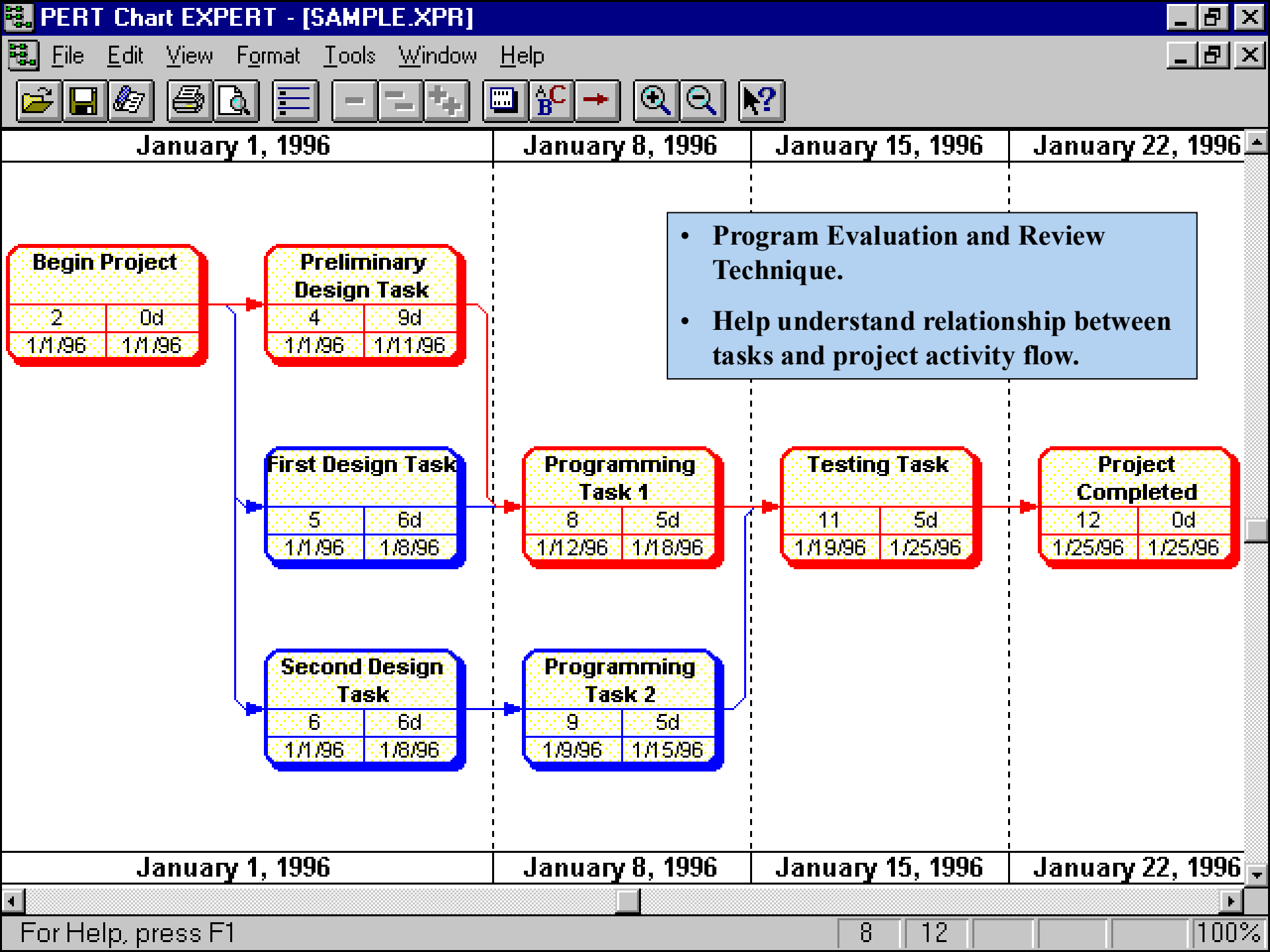
* Immediate preceding activity, ** In days

PERT Estimation Technique

What is a PERT?

Program Evaluation & Review Technique or PERT

- Identify the tasks (or activities) required to complete the given project
  - Use the WBS
- List the activities in a structured fashion, along with their interdependencies
  - Use a Gantt Chart
  - Precedence table
- A "network" of the activities and their dependencies is drawn up. In MS Project, PERT is called **Network Diagram**
- Each event may be represented by a node
- Before any activity can begin, all its predecessor activities must have been completed


PERT Chart EXPERT - [SAMPLE.XPR]
File Edit View Format Tools Window Help
January 1, 1996
January 8, 1996
January 15, 1996
January 22, 1996
Begin Project
2 0d
1/1/96 1/1/96
Preliminary Design Task
4 9d
1/1/96 1/11/96
First Design Task
5 6d
1/1/96 1/8/96
Second Design Task
6 6d
1/1/96 1/8/96
Programming Task 1
8 5d
1/12/96 1/18/96
Programming Task 2
9 5d
1/9/96 1/15/96
Testing Task
11 5d
1/19/96 1/25/96
Project Completed
12 0d
1/25/96 1/25/96
• Program Evaluation and Review Technique.
• Help understand relationship between tasks and project activity flow.
January 1, 1996
January 8, 1996
January 15, 1996
January 22, 1996
For Help, press F1
8
12
100%

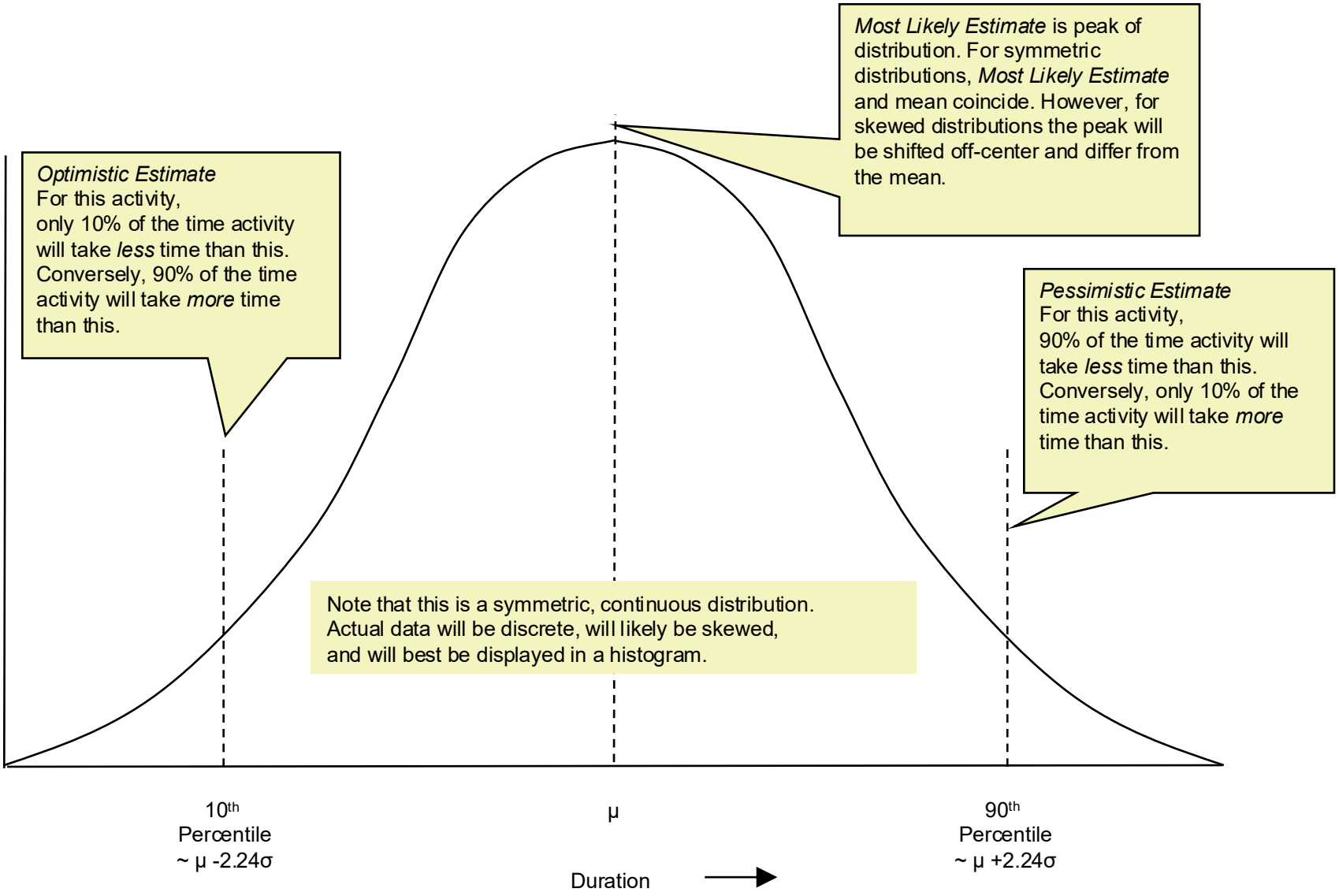
PERT estimation technique

- Most widely used three-point estimate is PERT (*Program Evaluation and Review Technique*)
- Defines three estimate points as:
  - *Most likely*: estimate that occurs with greatest frequency
  - *Optimistic*: shortest duration, taken as 10th percentile value
  - *Pessimistic*: longest duration, taken as 90th percentile value
  - PERT activity duration estimate $\boldsymbol{T_E}$ and its standard deviation (s_E or σ) are calculated according to:

$$T_E = (E_{optimistic} + 4E_{most_likely} + E_{pessimistic}) / 6$$

$$s_E = (E_{pessimistic} - E_{optimistic}) / 6$$

Duration estimates distribution



How to calculate PERT estimates

- This is for statistics majors and rarely used except when analyzing data.
- Calculate the sample mean: $\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$
- Calculate the sample standard deviation: $s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2}$
- Use *Chebyshev's rule* to approximate optimistic and pessimistic estimates:
  - At least $(1 - 1/k^2) \cdot 100\%$ of the values are within k standard deviations from the mean
- So, solving for k when this equation equals 80% (for the 20% outside the 10 and 90 percentiles), we get approximately:

$$Optimistic = \bar{x} - 2.24s \qquad Pessimistic = \bar{x} + 2.24s$$

PERT calculation example

You have the following **estimates for an activity**, in **staff-hours**:

24, 24, 24, 40, 48, 48

Calculate the **sample mean**:

$$\bar{x} = \frac{1}{6}(24+24+24+40+48+48)$$

$$\bar{x} = \frac{1}{6}(208) = 34.7$$

Calculate the **sample standard deviation** (s or σ):

$$s = \sqrt{\frac{1}{6-1}\sum_{i=1}^{6}(x_i - 34.7)^2}$$

$$s = \sqrt{\frac{1}{5}((24-34.7)^2 + (24-34.7)^2 + (24-34.7)^2 + (40-34.7)^2 + (48-34.7)^2 + (48-34.7)^2)}$$

$$s = \sqrt{(0.2)(114.5+114.5+114.5+28.1+176.9+176.9)}$$

$$s = \sqrt{0.2 \times 725.4}$$

$$s = \sqrt{145.1}$$

$$s \approx 12$$

PERT calculation example

Now, calculate the **PERT estimates** :

$$Optimistic = \bar{x} - 2.24s = 34.7 - (2.24 \times 12)$$

$$Optimistic = 34.7 - 26.9 = 7.8$$

$$Most_Likely = 24$$

$$Pessimistic = \bar{x} + 2.24s$$

$$Pessimistic = 34.7 + (2.24 \times 12) = 34.7 + 26.9 = 61.6$$

Finally, the **PERT estimated activity duration** is:

$$T_E = (E_{optimistic} + 4E_{most_likely} + E_{pessimistic}) / 6$$

$$T_E = (7.8 + (4 \times 24) + 61.6) / 6 = 165.4 / 6$$

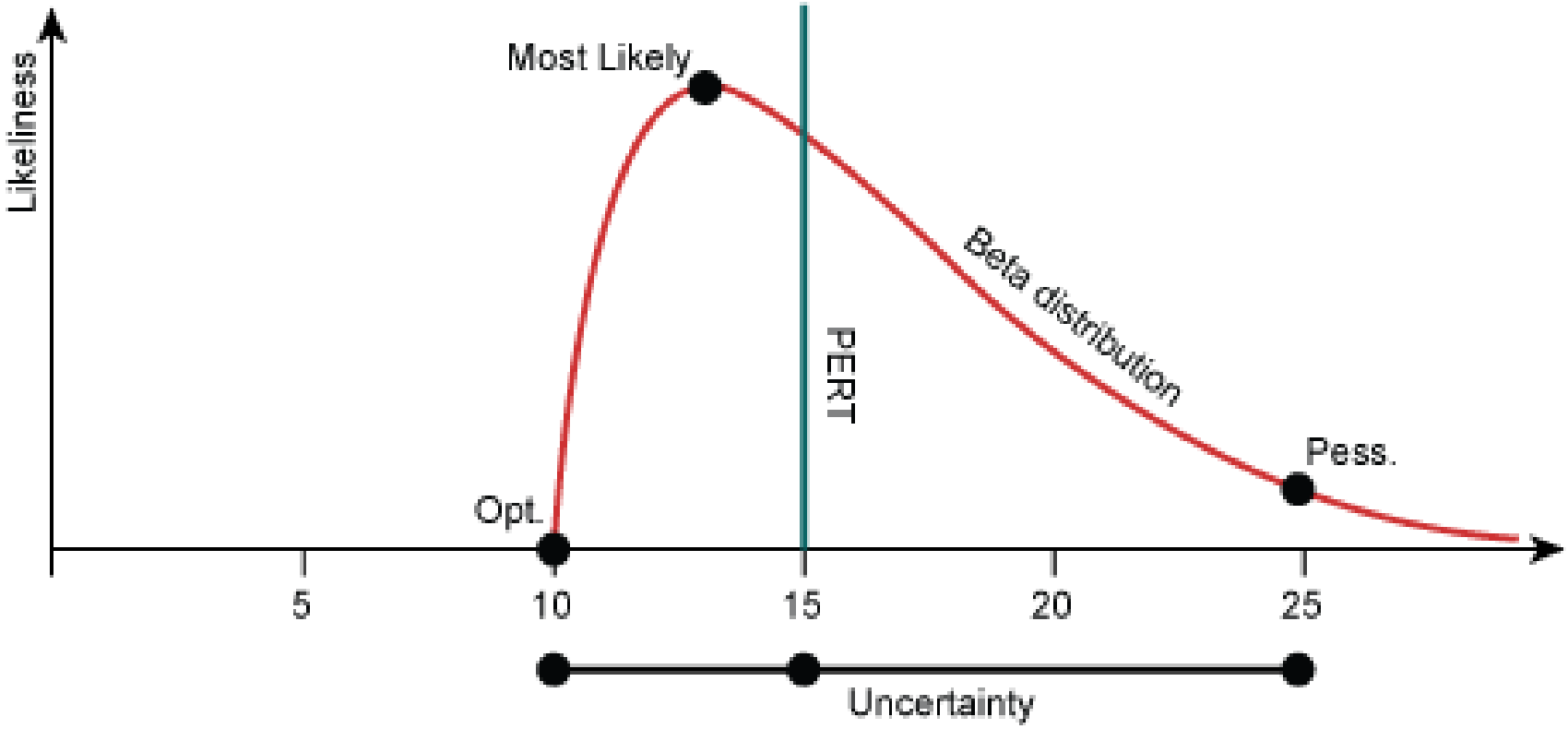
$$T_E = 27.6$$

$$s_E = (E_{pessimistic} - E_{optimistic}) / 6 = (61.6 - 7.8) / 6$$

$$s_E = 8.97 \approx 9$$

PERT calculation

Beta Distribution



Project Planning Tools

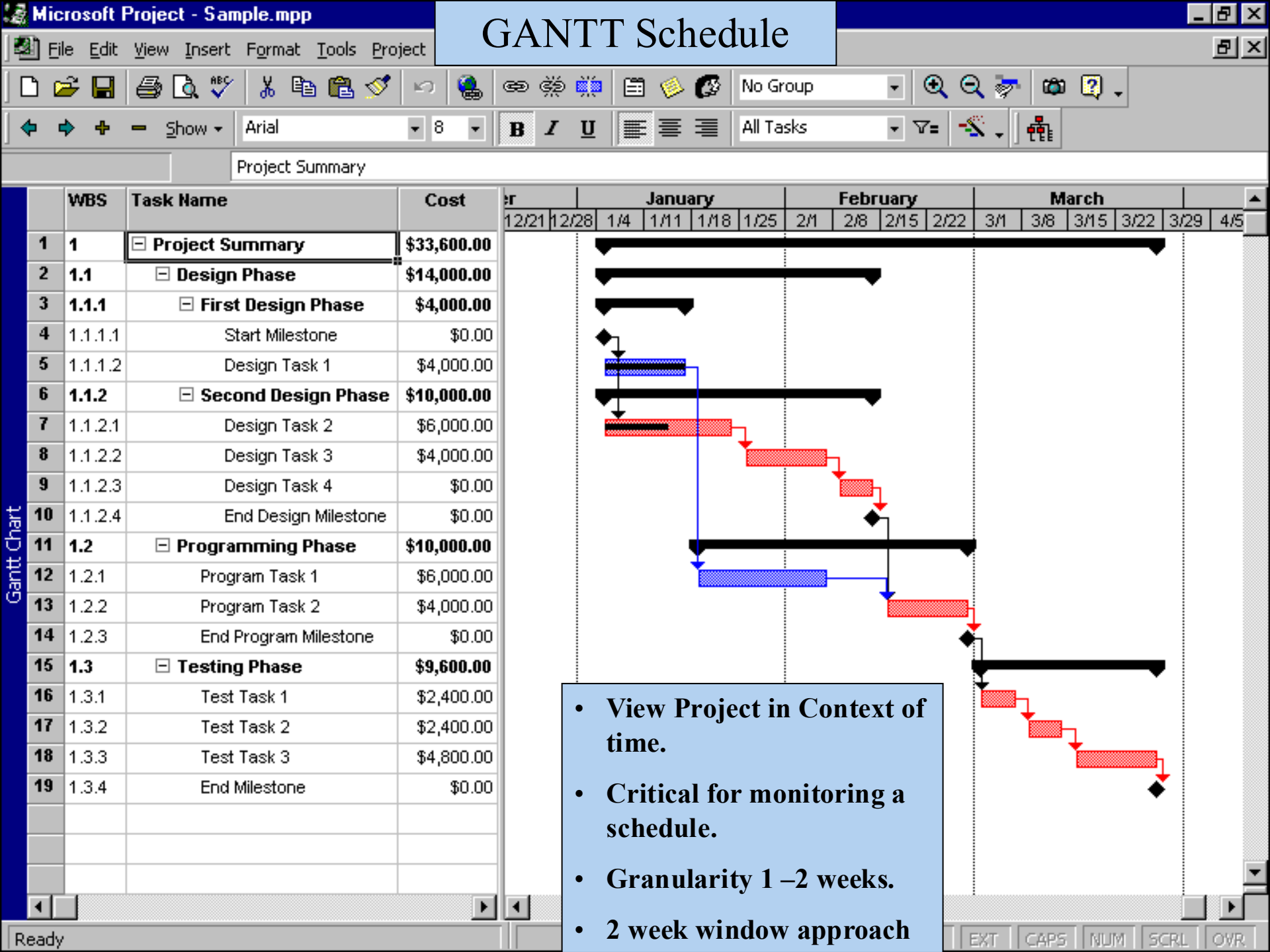
Gantt Chart

What is a Gantt chart?

A **Gantt Chart** (named for Henry Laurence Gantt) consists of a table of project task information and a bar chart that graphically displays project schedule, depicting progress in relation to time and often used in planning and tracking a project.

- Horizontal bar chart format, with bars representing the phases and activities of the WBS
- Time extends along the horizontal axis
- Able to show planned and actual progress on tasks as well as task dependencies
- Effective communication tool but with limitations
  - Very low information density: lots of wasted space
  - Not very useful for large projects

GANTT Schedule

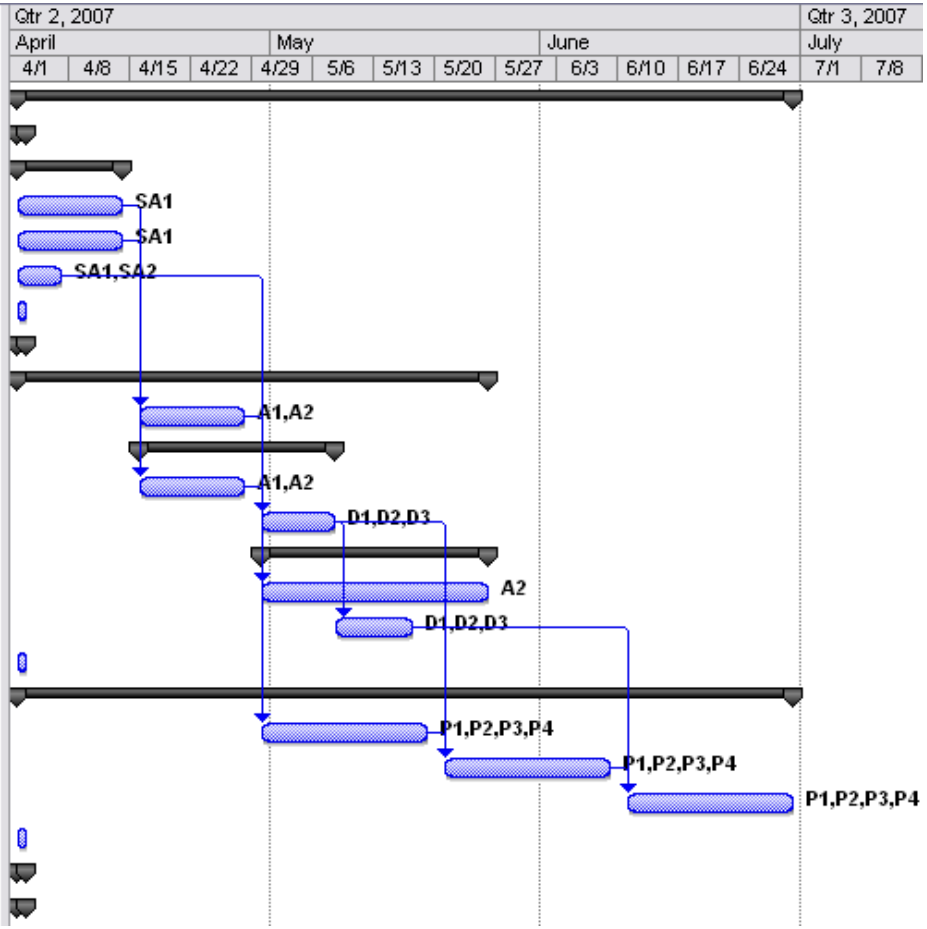


	WBS	Task Name	Cost
1	**1**	**Project Summary**	**$33,600.00**
2	**1.1**	**Design Phase**	**$14,000.00**
3	**1.1.1**	**First Design Phase**	**$4,000.00**
4	1.1.1.1	Start Milestone	$0.00
5	1.1.1.2	Design Task 1	$4,000.00
6	**1.1.2**	**Second Design Phase**	**$10,000.00**
7	1.1.2.1	Design Task 2	$6,000.00
8	1.1.2.2	Design Task 3	$4,000.00
9	1.1.2.3	Design Task 4	$0.00
10	1.1.2.4	End Design Milestone	$0.00
11	**1.2**	**Programming Phase**	**$10,000.00**
12	1.2.1	Program Task 1	$6,000.00
13	1.2.2	Program Task 2	$4,000.00
14	1.2.3	End Program Milestone	$0.00
15	**1.3**	**Testing Phase**	**$9,600.00**
16	1.3.1	Test Task 1	$2,400.00
17	1.3.2	Test Task 2	$2,400.00
18	1.3.3	Test Task 3	$4,800.00
19	1.3.4	End Milestone	$0.00

- **View Project in Context of time.**
- **Critical for monitoring a schedule.**
- **Granularity 1 –2 weeks.**
- **2 week window approach**

Sample evolutionary Gantt chart

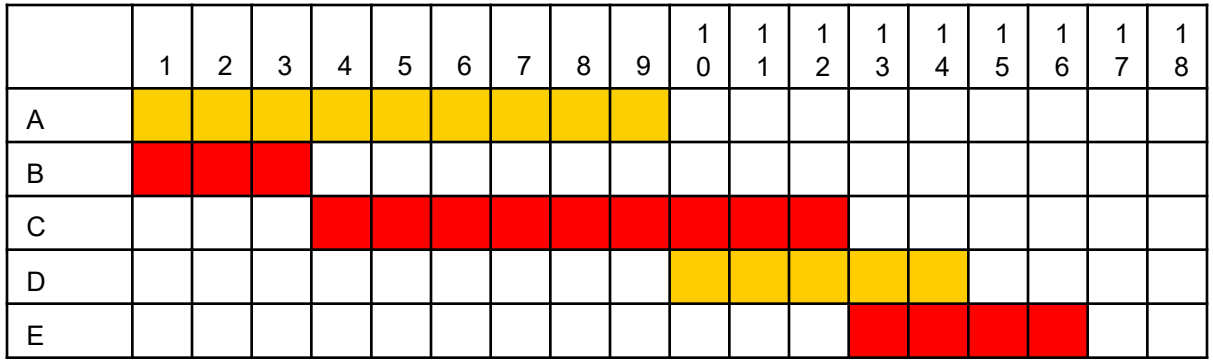
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36	**1.7**	**+ Deployment**	



Gantt Example:

Suppose a project comprises five activities: A,B,C,D, and E. A and B have no preceding activities, but activity C requires that activity B must be completed before it can begin. Activity D cannot start until both activities A and B are complete. Activity E requires activities A and C to be completed before it can start. If the activity times are A: 9 days; B: 3 days; C: 9 days; D: 5 days; and E: 4 days, determine the shortest time necessary to complete this project.

Identify those activities which are critical in terms of completing the project in the shortest possible time. [**Critical path]**



Time is 16 days
Critical path is
B, C, E